Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 15: Approved Solutions

Relations and Their Properties

Covers Chartrand & Zhang, Section 5.1

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Prove with an argument, disprove with names

Each property is a universal statement

	To prove	To disprove
Reflexive	argue for arbitrary a	name one a with $a \not R\, a$
Symmetric	argue for arbitrary a, b	name a, b: $a\, R\, b$, $b \not R\, a$
Transitive	argue for arbitrary a, b, c	name a, b, c

A disproof of transitivity needs *three* named elements, and you must state that the first two links hold and the third fails.

Three mistakes to expect

1. **"It's not transitive" with no witnesses.** Name the elements.
2. **Assuming the properties come as a package.** They are independent.
3. **Skipping edge cases.** Zero, equal elements, and the empty set break more relations than anything else.

Part II: Practice Problems

Section A — Relations given as lists

1. Let $A = \{1, 2, 3\}$. For each relation, state which of reflexive, symmetric, transitive it has, with a witness for each failure.

(a) $R_1 = \{(1,1), (2,2), (3,3), (1,3), (3,1)\}$

(b) $R_2 = \{(1,2), (2,3), (1,3)\}$

(c) $R_3 = \{(1,1), (2,2), (3,3), (1,2), (2,3)\}$

(d) $R_4 = \emptyset$

Approved solution

(a) All three. The diagonal is complete; $(1,3)$ and $(3,1)$ appear together; the only chains are within $\{1,3\}$ and close. It is an equivalence relation.
(b) Transitive only. Not reflexive: $1 \not R_2\ 1$. Not symmetric: $(1,2) \in R_2$ but $(2,1) \notin R_2$. Transitive: the only chain is $(1,2), (2,3)$, and $(1,3)$ is present.
(c) Reflexive only. Not symmetric: $(1,2)$ without $(2,1)$. Not transitive: $(1,2)$ and $(2,3)$ are present but $(1,3)$ is not.
(d) Symmetric and transitive, not reflexive. The empty relation has no pair to violate symmetry or transitivity, so both hold **vacuously**. It fails reflexivity because $(1,1) \notin \emptyset$.

Part (d) is the one worth arguing about. Both properties are implications, and an implication with no instances of its hypothesis is vacuously true — Lecture 4, still paying rent. Students who answer "it has no properties" have not applied the definitions.

2. Let $A = \{1, 2, 3, 4\}$. Give an example of a relation on A that is

(a) symmetric but not reflexive;

(b) reflexive and symmetric but not transitive;

(c) reflexive and transitive but not symmetric.

Approved solution

(a) $\{(1,2), (2,1)\}$. Symmetric, but $1 \not R\ 1$.
(b) $\{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (2,3), (3,2)\}$. Reflexive and symmetric, but $1\ R\ 2$ and $2\ R\ 3$ while $(1,3)$ is absent.
(c) $\leq$ restricted to A, that is $\{(a,b) \in A \times A : a \leq b\}$. Reflexive and transitive; not symmetric, since $1 \leq 2$ but $2 \not\leq 1$.

The point of the problem is that the three properties are independent: any two can hold while the third fails. Answers vary; check each against the definitions rather than against these.

Section B — Relations on $\mathbb{Z}$ and $\mathbb{R}$

3. For each relation on $\mathbb{Z}$, determine which properties hold. Give a witness for each failure.

(a) $a\ R\ b$ if $a + b$ is odd.

(b) $a\ R\ b$ if a divides b.

(c) $a\ R\ b$ if $a - b$ is divisible by 3.

(d) $a\ R\ b$ if $|a - b| \leq 1$.

Approved solution

(a) Symmetric only. Not reflexive: $a + a = 2a$ is even, so $a \not R\ a$ for every a. Symmetric since $a + b = b + a$. Not transitive: $1\ R\ 2$ and $2\ R\ 3$, but $1 + 3 = 4$ is even so $1 \not R\ 3$.
(b) Reflexive and transitive. Every a divides itself. If $a \mid b$ and $b \mid c$ then $a \mid c$. Not symmetric: $2 \mid 4$ but $4 \nmid 2$.
(c) All three — an equivalence relation. This is congruence modulo 3.
(d) Reflexive and symmetric, not transitive. $|a - a| = 0 \leq 1$; the condition is symmetric in a and b. Witness for transitivity: $1\ R\ 2$ and $2\ R\ 3$, but $|1 - 3| = 2 > 1$.

Part (d) is the "close enough" relation, and its failure is the reason "nearly equal" is not an equivalence relation: small differences accumulate. That matters in floating point, where testing approximate equality is famously not transitive.

4. A relation is defined on $\mathbb{R}$ by $x\ R\ y$ if $x - y \geq 0$.

(a) Give two reals related by R and two that are not.

(b) Determine which properties R has, with witnesses.

Approved solution

(a) $5\ R\ 3$ since $5 - 3 = 2 \geq 0$. And $3 \not R\ 5$ since $3 - 5 = -2 < 0$.
(b) R is just $\geq$. **Reflexive:** $x - x = 0 \geq 0$. **Not symmetric:** $5\ R\ 3$ but $3 \not R\ 5$. **Transitive:** if $x - y \geq 0$ and $y - z \geq 0$, adding gives $x - z \geq 0$.
The addition step is the same move as in Wednesday's transitivity proof — combine the two hypotheses and let the middle term cancel.

Section C — Counting

5. Let $|A| = n$.

(a) How many relations are there on A?

(b) How many are reflexive?

(c) For $n = 3$, give both numbers.

Approved solution

(a) A relation is a subset of $A \times A$, which has n^2 elements. By Theorem 2.17 the number of subsets is 2^{n^2}.
(b) A reflexive relation must contain all n diagonal pairs (a, a) and may contain any of the remaining $n^2 - n$ pairs freely. So there are $2^{n^2 - n}$.
(c) For $n = 3$: $2^9 = 512$ relations, of which $2^6 = 64$ are reflexive.

Both counts are the power-set theorem applied to $A \times A$, with part (b) fixing some choices in advance. That is the same reasoning as Lecture 1's password count with certain characters forced.

Section D — Writing

6. In five to eight complete sentences, explain why disproving that a relation is transitive requires naming three elements, while disproving reflexivity requires only one. Illustrate each with a specific relation.

Approved solution

Model answer. Each property is a universal statement, so its negation is existential and disproving it means exhibiting a witness — but the witnesses have different shapes because the statements quantify over different numbers of variables. Reflexivity says that for every a, $a \mathrel{R} a$, which involves a single variable, so its negation asks for a single element failing the condition. For instance, $ab > 0$ on $\mathbb{Z}$ is not reflexive, and $a = 0$ alone shows it. Transitivity says that for all a, b, c, if $a \mathrel{R} b$ and $b \mathrel{R} c$ then $a \mathrel{R} c$, which quantifies over three variables, so a witness must supply all three. Moreover the inner statement is an implication, so the witness must make the hypothesis true and the conclusion false: both links present and the shortcut missing. For $|a - b| \leq 1$ on $\mathbb{Z}$, the triple $1, 2, 3$ works, since $1 \mathrel{R} 2$ and $2 \mathrel{R} 3$ but $|1 - 3| = 2 > 1$.

Full credit requires both the variable count *and* the observation that a transitivity witness must satisfy the hypothesis. A student who names three elements failing the first link has not produced a counterexample — that is the most common error.

Section E — Challenge (optional)

7. Prove that if R is symmetric and transitive and *every element of A is related to at least one element*, then R is reflexive. Then explain why the extra hypothesis cannot be dropped.

Approved solution

Proof. Let $a \in A$. By hypothesis there is some $b \in A$ with $a \mathrel{R} b$. By symmetry, $b \mathrel{R} a$. By transitivity applied to $a \mathrel{R} b$ and $b \mathrel{R} a$, we get $a \mathrel{R} a$. Since a was arbitrary, R is reflexive. ■

Why the hypothesis is needed. Take $A = \{1, 2\}$ and $R = \emptyset$. This is vacuously symmetric and transitive, but not reflexive — and no element is related to anything, so the extra hypothesis fails. A less degenerate example: $A = \{1, 2\}$ with $R = \{(1, 1)\}$ is symmetric and transitive but $2 \not\mathrel{R} 2$.

Why this matters. There is a well-known bad argument claiming reflexivity follows from symmetry and transitivity alone: "$a \mathrel{R} b$, so $b \mathrel{R} a$, so $a \mathrel{R} a$." The flaw is the opening — it assumes a is related to *something*, which is exactly the hypothesis we had to add. Spotting a missing quantifier in a plausible argument is a skill worth having.

Instructor notes

- **Problem 1(d) — the empty relation** — is the best item here. Symmetric and transitive vacuously, not reflexive. It rewards students who apply definitions rather than intuition.
- **Problem 3(d)** is "close enough," whose non-transitivity explains why approximate float equality misbehaves. Worth mentioning to CS students explicitly.
- **Problem 7 is the one to put on the board.** The bad argument that reflexivity follows from the other two properties is genuinely tempting, and locating the missing quantifier is excellent practice.
- **Problem 6** exposes the most common disproof error: naming three elements that do not satisfy the hypothesis. Watch for it on Homework 7.
- Reserved for Homework 7: Section 5.1 Exercises 8 and 10 (even). Odd ones have printed answers

and are self-check only.